

Dear Editor,

Please consider my manuscript entitled:

"Machine Learning Assisted Analysis of a Static Relativistic Gravitational Potential: A Comparative Study with Newtonian Approximation and Error Quantification"

for publication as a Research Article in the European Journal of Physics.

In this work, I present a computational study of a static relativistic gravitational potential using a machine learning-assisted approach. I compare relativistic predictions with the Newtonian approximation and quantify the regimes where classical approximations remain valid or begin to deviate.

The manuscript combines concepts from gravitational physics, numerical analysis, and scientific machine learning to provide a computational framework for analysing relativistic corrections and approximation errors.

I confirm that this manuscript represents original work, has not been published previously, and is not under consideration by any other journal. I have read and agree to the policies of the journal.

I believe this work will be of interest to readers of the European Journal of Physics, particularly those interested in computational approaches to physics modelling and the application of modern computational methods in physics.

Thank you for your consideration.

Sincerely,

Aayush Kumar
Corresponding Author